Plant Safety Manual — Rotating Equipment

Applies to: Pumps, Compressors, Motors — Sections P-100 to P-150

1. General Safety Rules

No physical inspection or maintenance work may be performed on rotating equipment while it is energized and running, except for non-contact measurements explicitly permitted by procedure (e.g. thermal imaging, vibration measurement with a handheld probe). Before any hands-on work, equipment must be electrically isolated and locked out (LOTO) by an authorized technician, and residual pressure in the system must be verified as zero before opening any casing.

2. Personal Protective Equipment (PPE)

Minimum PPE for work in the pump area: safety glasses, hearing protection (area noise level exceeds 85 dB), steel-toe boots, and cut-resistant gloves. Hot surfaces (bearing housings, motor casings) may exceed 70 C during normal operation; thermal gloves are required for any contact measurement close to these surfaces.

3. Safety-Critical Thresholds

Condition	Threshold	Required Action
Bearing temperature	> 90 C	Immediate shutdown
Vibration (RMS velocity)	> 11 mm/s	Immediate shutdown
Motor winding temperature	> 165 C	Immediate shutdown
Seal chamber pressure loss	< 0.5 bar sustained	Shutdown, inspect seal
Audible unusual noise (grinding/knocking)	Any occurrence	Reduce load, notify supervisor

4. Work Authorization

Any corrective action that requires stopping production equipment (e.g. taking P-102 offline for bearing replacement) must be authorized by the shift supervisor before a work order is created or executed. Routine, non-invasive diagnostic actions (reading sensor data, reviewing maintenance history) do not require prior authorization.

5. Emergency Contacts

Shift supervisor: extension 4501. Maintenance control room: extension 4510. Plant safety officer: extension 4520. In case of fire or uncontrolled leak, activate the nearest emergency stop and follow the plant evacuation plan posted at each access point.